



GOSFORD HIGH SCHOOL

2011 TRIAL HSC EXAMINATION

EXTENSION 2 MATHEMATICS

General Instructions:

- Reading time: 5 minutes.
- Working time: 3 hours
- Write using black or blue pen.
- Board-approved calculators may be used.
- Each question should be started on a separate writing booklet.
- All necessary working should be shown in every question.

Total marks: - 120

Attempt all Questions 1- 8.

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{dx}{\sqrt{9x^2-1}}$. (2)

(b) Find $\int \frac{dx}{\sqrt{4x-x^2}}$. (2)

(c) Evaluate $\int_0^\pi x \sin x \, dx$. (3)

(d) Find $\int \cos^5 x \sin^2 x \, dx$. (4)

(e) Use the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$. (4)

Question 2 (15 marks) Use a SEPARATE writing booklet.

(a) If $z = 2 + i$ and $\omega = 1 - 3i$ find in the form $x + iy$

(i) z^2 . (1)

(ii) $z\bar{\omega}$. (1)

(iii) $\frac{z}{\omega}$. (1)

(b)

(i) Express $z = 1 + \sqrt{3}i$ in modulus-argument form. (2)

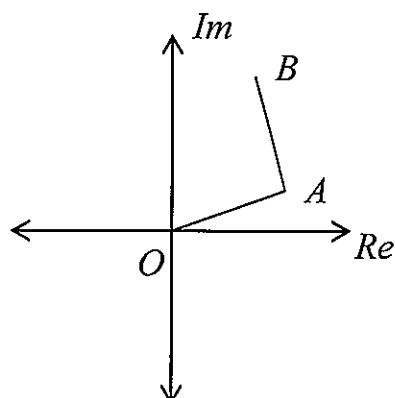
(ii) Show that $(1 + \sqrt{3}i)^6$ is a real number. (2)

(c) For the complex number $z = x + iy$, where x and y are real numbers, find and clearly sketch the curve on an Argand diagram for which

(i) $|z + \bar{z}| \leq 2$. (2)

(ii) $\operatorname{Re}(z^2 - 4) = 0$. (3)

(d) The point A in the Argand diagram below represents the complex number $z = a + ib$. The point B represents the complex number $2 + 5i$.

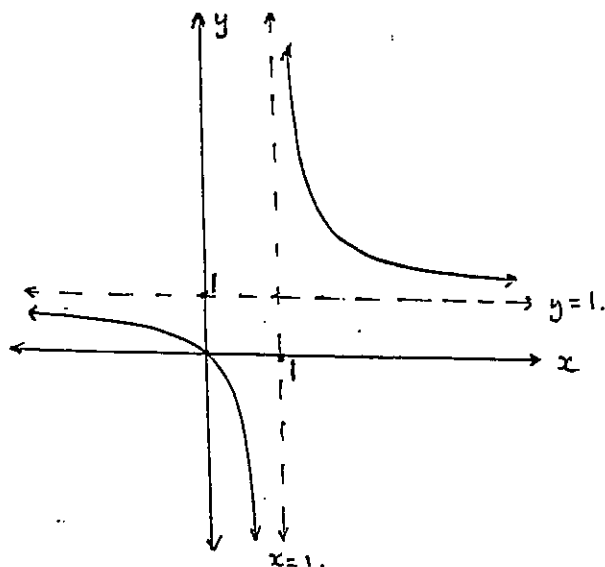


NOT TO SCALE

If the complex number represented by the point C is such that OABC is a square, find C in terms of a and b and hence evaluate a and b . (3)

Question 3 (15 marks) Use a SEPARATE writing booklet.

(a) The function defined by $y = f(x)$ is drawn below.



Draw separate one-third page sketches of

(i) $y = f(x)$ and $y = f(-x)$. (2)

(ii) $y = f(x)$ and $y = \frac{1}{f(x)}$. (2)

(iii) $y = f(x)$ and $|y| = f(x)$. (2)

(iv) $y = f(x)$ and $y^2 = f(x)$. (3)

(b) The equation of a curve is $4x^2 + xy + y^2 = 10$. Find the equation of the tangent to the curve at the point (1,2) on it. (3)

(c) Find the number of different ways of arranging any 4 of the letters from the word EXERCISES. (3)

Question 4 (15 marks) Use a SEPARATE writing booklet.

(a) When a polynomial $P(x)$ is divided by $(x - 3)$ the remainder is 10 and when $P(x)$ is divided by $(x - 4)$ the remainder is 13. Determine the remainder when $P(x)$ is divided by $(x - 3)(x - 4)$. (2)

(b) If α, β and γ are the roots of the equation $x^3 - 7x^2 - 7 = 0$ find the equations whose roots are

(i) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$. (2)

(ii) $\alpha^2, \beta^2, \gamma^2$. (2)

(c)

(i) Express $\frac{2}{x^3+2x}$ in the form $\frac{A}{x} + \frac{Bx+C}{x^2+2}$. (2)

(ii) Show that $\int_1^2 \frac{2}{x^3+2x} dx = \frac{1}{2} \ln 2$. (2)

(d) Consider the equation $z^4 + pz^3 + qz + r = 0$, where p, q & r are real numbers. The sum of the roots of this equation is 6 more than the product of the roots.

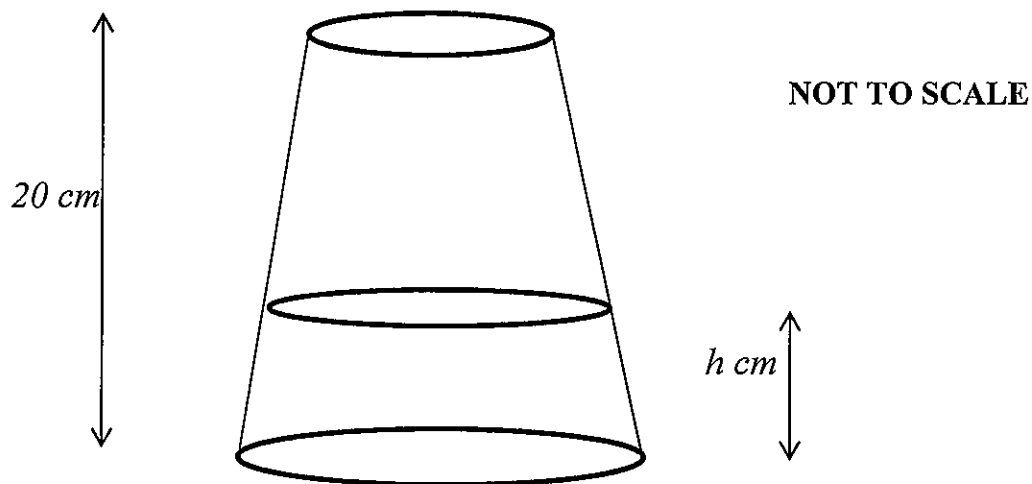
If $1 + i$ is a root of the equation, find

(i) p, q & r . (3)

(ii) all the roots of the equation. (2)

Question 5 (15 marks) Use a SEPARATE writing booklet.

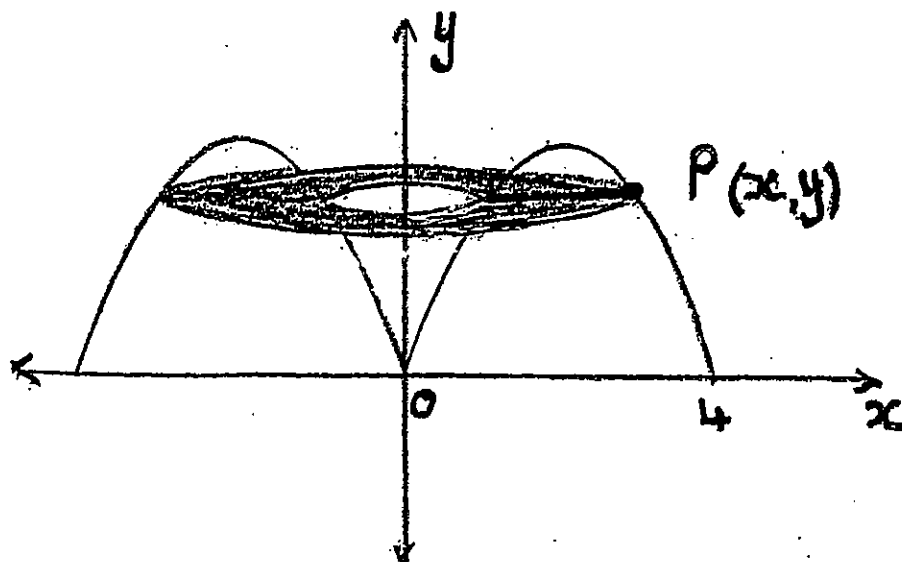
- (a) The region bounded by the x-axis and the curve $y = -2 + 3x - x^2$ is rotated about the line $x = 3$ to form a solid. Use the method of cylindrical shells to find the volume of the solid formed. (5)
- (b) The diagram below shows the frustrum of a right cone. (A frustrum of a cone is a cone with its top cut off.) The height of the frustrum is 20 cm and the radii of the base and the top are 15 cm and 10 cm respectively.



A horizontal cross-section taken at height h cm is a circle of radius r units.

- (i) Show that $r = 15 - \frac{h}{4}$. (2)
- (ii) Find the volume of the frustrum. (3)

- (c) The region bounded by $y = 4x - x^2$ and the x-axis is rotated about the y-axis to form a solid of revolution. If a horizontal line is drawn from the point $P(x, y)$ on the curve, where $2 < x < 4$, to the y-axis it sweeps out an annulus.



- (i) Show that the area of the annulus is given by

$$A = \pi[4\sqrt{16 - 4y}]. \quad (3)$$

- (ii) Hence find the volume of the solid. (2)

Question 6 (15 marks) Use a SEPARATE writing booklet.

(a) Consider the ellipse $\mathcal{E}$, with equation $\frac{x^2}{100} + \frac{y^2}{64} = 1$.

(i) Calculate the eccentricity of $\mathcal{E}$. (1)

(ii) Find the coordinates of the foci and the equations of the directrices of $\mathcal{E}$. (2)

(iii) Show that the equation of the tangent at the point $P(x_0, y_0)$ on $\mathcal{E}$ is

$$\frac{x_0 x}{100} + \frac{y_0 y}{64} = 1. \quad (3)$$

(b) A conic is a rectangular hyperbola with eccentricity $\sqrt{2}$, focus $(2, 0)$ and directrix $x = 1$.

(i) Find the equation of this hyperbola. (1)

(ii) Sketch this hyperbola clearly showing the asymptotes and vertices. (1)

(iii) Show that the equation of the normal at the point $P(a \sec \theta, a \tan \theta)$ is $x \tan \theta + y \sec \theta = 2\sqrt{2} \sec \theta \tan \theta$. (3)

(iv) This normal meets the x-axis at $Q(X, 0)$ and the y-axis at $R(0, Y)$.

If T is the point (X, Y) , find the locus of T and describe this locus geometrically. (4)

Question 7 (15 marks) Use a SEPARATE writing booklet.

(a) A particle of unit mass is projected vertically upwards from ground level with initial speed U . Assume that air resistance is kv , where v is the particle's speed and k is a positive constant. We wish to consider the particle's motion as it falls back to ground level. Let y be the displacement of the particle measured vertically downwards from the point of maximum height, t be the time elapsed after the particle has reached maximum height, and g be the acceleration due to gravity.

(i) Explain why $v(0) = 0$ and $\frac{dv}{dt} = g - kv$ while the particle is in motion. (1)

(ii) Deduce that $v = \frac{g}{k} (1 - e^{-kt})$ for $t \geq 0$. (3)

(iii) By writing $\frac{dv}{dt} = v \frac{dv}{dy}$, deduce from part (i) that

$$\frac{g}{k} \log_e \left(\frac{g - kv}{g} \right) + v = -ky. \quad (3)$$

(iv) Using parts (ii) and (iii) deduce that $t = \frac{v + ky}{g}$. (2)

(v) Given that the particle reaches a maximum height

$$h = \frac{1}{k} \left[U - \frac{g}{k} \log_e \left(\frac{g + kU}{g} \right) \right] \text{ in time } t_h = \frac{1}{k} \log_e \left(\frac{g + kU}{g} \right), \text{ deduce}$$

that the total time T that the particle is in the air is $T = \frac{U + V}{g}$, where V is the final speed of the particle when it returns to ground level. (1)

(b) If $I_n = \int_0^1 (x^2 - 1)^n dx$, $n = 0, 1, 2, \dots$

(i) Show that $I_0 = 1$. (1)

(ii) Prove that $I_n = \frac{-2n}{2n+1} I_{n-1}$. (3)

(iii) Hence evaluate $\int_0^1 (x^2 - 1)^4 dx$. (1)

Question 8 (15 marks) Use a SEPARATE writing booklet.

(a)

(i) Use DeMoivre's Theorem to express $\cos 4\theta$ and $\sin 4\theta$ in powers of $\cos \theta$ and $\sin \theta$. Hence express $\tan 4\theta$ as a rational function of t , where $t = \tan \theta$. (4)

(ii) Hence solve the equation $t^4 + 4t^3 - 6t^2 - 4t + 1 = 0$. (3)

(b) A particle is projected from the origin with an initial velocity of $V \text{ ms}^{-1}$ at an angle of α to the horizontal.

(i) Show that the maximum range on the horizontal plane is $\frac{V^2}{g}$ when $\alpha = \frac{\pi}{4}$. (4)

(ii) The particle is now to hit a target which is h metres above its horizontal position when the maximum range in part (i) is reached. If the angle of projection α remains the same, show that the initial velocity must be increased from $V \text{ ms}^{-1}$ to $\frac{V^2}{\sqrt{V^2 - gh}} \text{ ms}^{-1}$. (Air resistance is neglected). (4)

END OF PAPER

QUESTION 1.

GLENN,
PLEASE MARK ① & 2

$$\begin{aligned}
 a) \int \frac{dx}{\sqrt{9x^2 - 1}} &= \int \frac{dx}{3\sqrt{x^2 - \frac{1}{9}}} \\
 &= \frac{1}{3} \int \frac{dx}{\sqrt{x^2 - \frac{1}{9}}} \quad (2) \\
 &= \frac{1}{3} \ln \left| x + \sqrt{x^2 - \frac{1}{9}} \right| + C, \quad x > \frac{1}{3} > 0.
 \end{aligned}$$

$$\begin{aligned}
 b) \int \frac{dx}{\sqrt{4x - x^2}} &= \int \frac{dx}{\sqrt{4 - (x^2 - 4x + 4)}} \\
 &= \int \frac{dx}{\sqrt{4 - (x-2)^2}} \quad (2) \\
 &= \sin^{-1} \left[\frac{x-2}{2} \right] + C
 \end{aligned}$$

$$c) \int_0^{\pi} x \sin x \, dx$$

$$\begin{aligned}
 \text{Let } u &= x & v' &= \sin x \\
 u' &= 1 & v &= -\cos x
 \end{aligned}$$

$$\begin{aligned}
 I &= \left[-x \cos x \right]_0^{\pi} - \int_0^{\pi} -\cos x \, dx \\
 &= (-\pi \cos \pi - 0) + \left[\sin x \right]_0^{\pi} \quad (3)
 \end{aligned}$$

$$= \pi - 0 + \sin \pi - \sin 0$$

$$= \pi - 0 + 0 - 0$$

$$= \pi$$

$$d) \int \cos^5 x \sin^2 x \, dx$$

$$= \int \cos^4 x \sin^2 x \cos x \, dx$$

$$= \int (1 - \sin^2 x)^2 \sin^2 x \cos x \, dx$$

$$\text{Let } u = \sin x$$

$$du = \cos x \, dx$$

$$I = \int (1 - u^2)^2 u^2 \, du$$

$$= \int u^2 (1 - 2u^2 + u^4) \, du$$

$$= \int u^2 - 2u^4 + u^6 \, du$$

$$= \frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} + C$$

$$= \frac{\sin^3 x}{3} - \frac{2\sin^5 x}{5} + \frac{\sin^7 x}{7} + C$$

$$c) \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$$

$$\text{Let } t = \tan \frac{x}{2}$$

$$dx = \frac{2 \, dt}{1 + t^2}$$

$$\text{When } x=0, t=0$$

$$x = \frac{\pi}{2}, t=1$$

$$\cos x = \frac{1 - t^2}{1 + t^2}$$

$$I = \int_0^1 \frac{1}{2 + \frac{1 - t^2}{1 + t^2}} \cdot \frac{2 \, dt}{1 + t^2}$$

$$= \int_0^1 \frac{1}{\frac{2 + 2t^2 + 1 - t^2}{1 + t^2}} \cdot \frac{2 \, dt}{1 + t^2}$$

$$= \int_0^1 \frac{2}{3 + t^2} \, dt$$

$$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} - 0 \right) = \frac{\pi}{3\sqrt{3}}$$

QUESTION 2

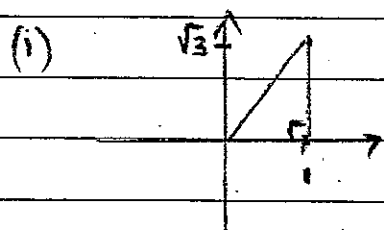
a) $z = 2+i$, $w = 1-3i$

$$\begin{aligned} \text{(i)} \quad z^2 &= (2+i)^2 \\ &= 4 + 4i + i^2 \\ &= 3 + 4i \end{aligned} \quad (1)$$

$$\begin{aligned} \text{(ii)} \quad z\bar{w} &= (2+i)(1+3i) \\ &= 2 + 6i + i + 3i^2 \\ &= -1 + 7i \end{aligned} \quad (1)$$

$$\begin{aligned} \text{(iii)} \quad \frac{z}{w} &= \frac{(2+i) \times (1+3i)}{(1-3i)(1+3i)} \\ &= \frac{-1+7i}{1-9i^2} \\ &= \frac{-1+7i}{10} \\ &= -\frac{1}{10} + \frac{7}{10}i \end{aligned} \quad (1)$$

b) $z = 1 + \sqrt{3}i$



$$\begin{aligned} |z| &= \sqrt{(\sqrt{3})^2 + 1^2} \\ &= 2 \\ \arg z &= \tan^{-1} \sqrt{3} \\ &= \frac{\pi}{3} \end{aligned}$$

$$\therefore z = 2 \operatorname{cis} \frac{\pi}{3} \quad (2)$$

$$\text{(iv)} \quad z^6 = 2^6 \operatorname{cis} 6 \frac{\pi}{3}$$

$$= 64 \operatorname{cis} 2\pi$$

$$= 64 (\cos 2\pi + i \sin 2\pi)$$

$$= 64 (1 + 0i)$$

$$= 64 \text{ which is real.} \quad (2)$$

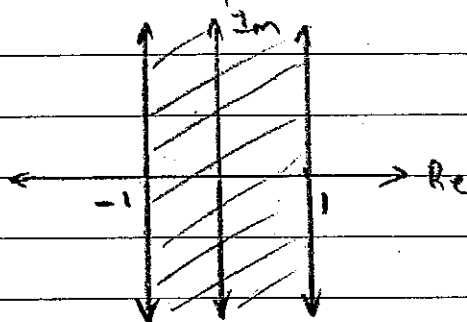
c)

(i) If $z = x + iy$, $\bar{z} = x - iy$
 $\therefore z + \bar{z} = 2x$

If $|z + \bar{z}| \leq 2$

$|2x| \leq 2$

$|x| \leq 1$



(2)

(iii) If $z = x + iy$
 $z^2 = (x + iy)^2$
 $= x^2 + 2xyi + i^2y^2$
 $= (x^2 - y^2) + 2xyi$

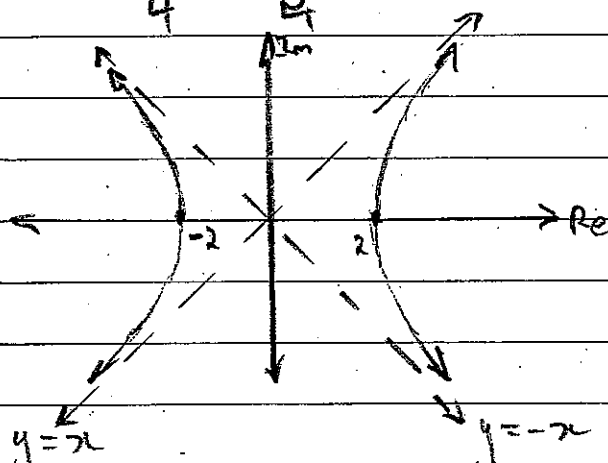
$\therefore \text{Re}(z^2 - 4) = x^2 - y^2 - 4$

If $\text{Re}(z^2 - 4) = 0$

$x^2 - y^2 - 4 = 0$

$x^2 - y^2 = 4$

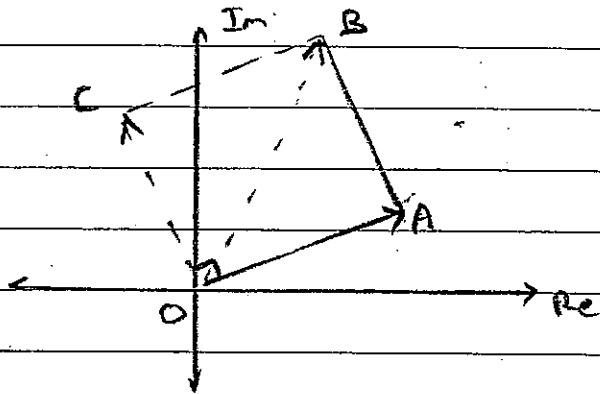
$\frac{x^2}{4} - \frac{y^2}{4} = 1$



This is a rect.
 hyperbola with
 asymptotes $y = \pm x$,
 x intercepts ± 2 .

(3)

d)



If A is $z = a + ib$
 C is $iz = i(a + ib)$
 $= ia + i^2 b$
 $\therefore C$ is $-b + ia$ (1)

$$\vec{OB} = \vec{OA} + \vec{OC}$$

$$\therefore 2 + 5i = a + ib + -b + ia$$

$$2 + 5i = a - b + i(a + b)$$

$$a - b = 2 \quad \text{--- (1)}$$

$$a + b = 5 \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2}$$

$$2a = 7$$

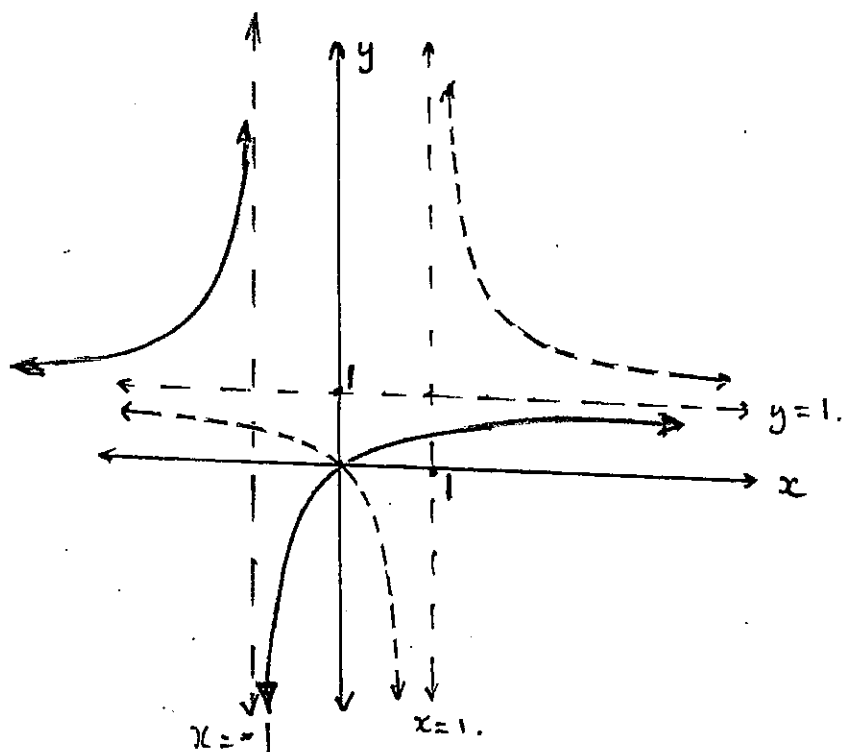
$$a = \frac{7}{2}$$

(2)

$$b = \frac{3}{2}$$

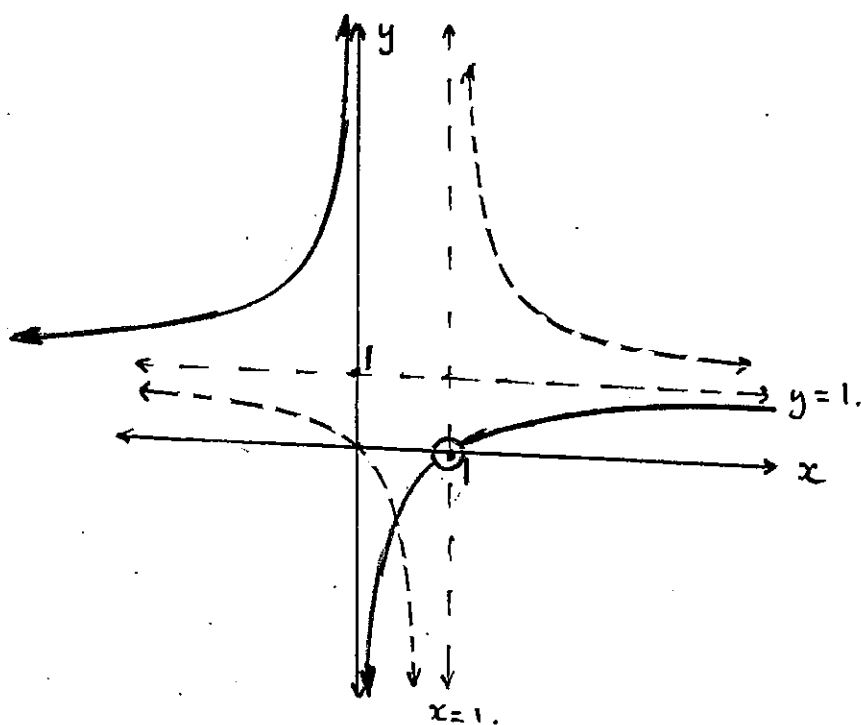
QUESTION 3.

a) (i)



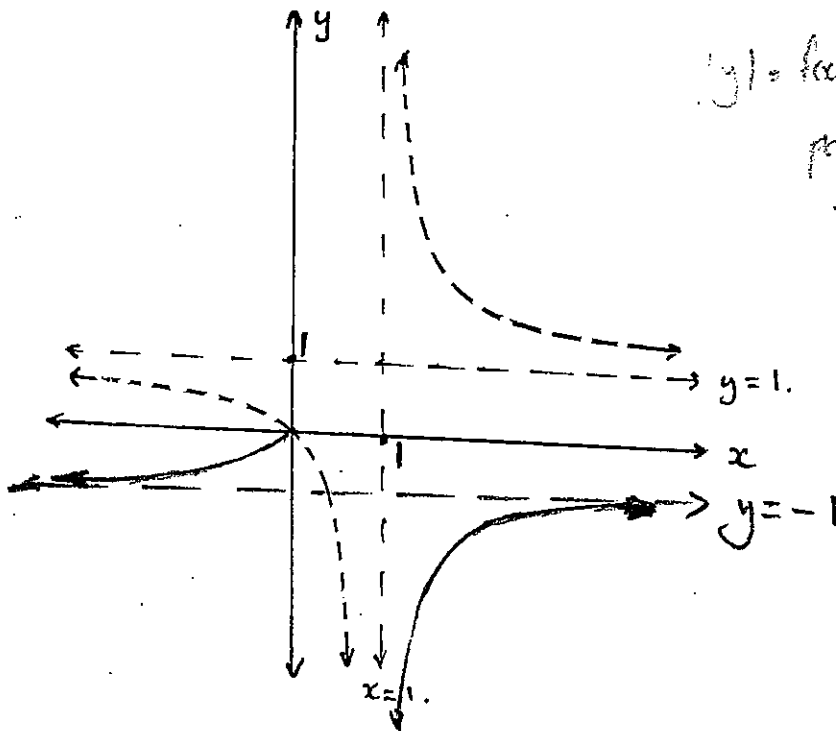
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(ii)



2

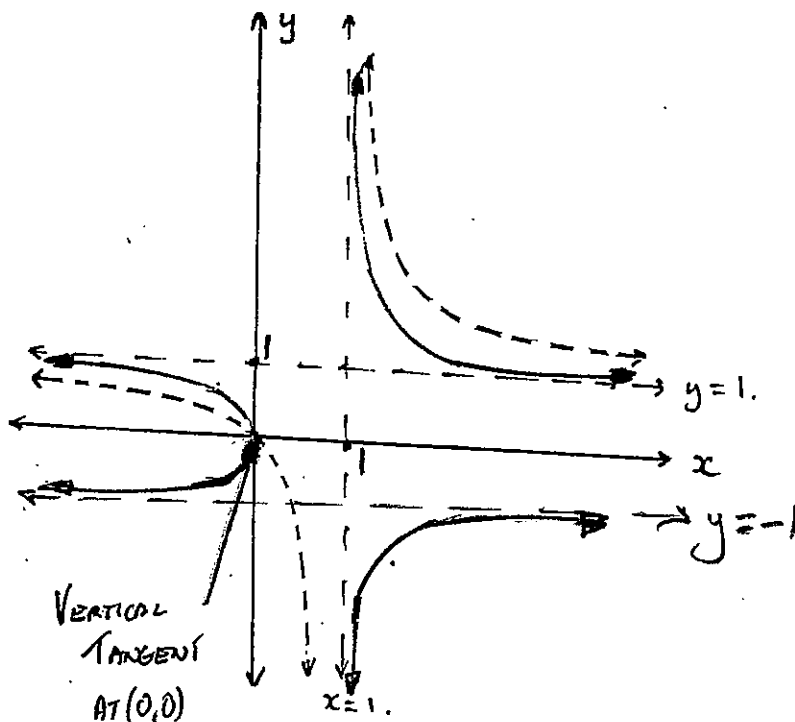
(III)



$y = f(x)$ consist of the part of $f(x)$ above the x-axis & its reflection in the x-axis

2

(IV)



$$y^2 = f(x)$$
$$y = \pm \sqrt{f(x)}$$

3

b) If $4x^2 + xy + y^2 = 10$
By implicit differentiation

$$8x + y \cdot 1 + x \cdot y' + 2y \cdot y' = 0$$

$$y'(x+2y) = -8x-y$$

$$y' = \frac{-8x-y}{x+2y}$$

When $x=1, y=2$

$$y' = \frac{-8-2}{1+4}$$

$$= -2$$

2

$\therefore$ Eqⁿ of tangent is

$$y-2 = -2(x-1)$$

$$y-2 = -2x+2$$

$$2x+y-4=0$$

c) Groups of 4 letters can be classified as

(i) 3 alike & 1 different

(ii) 2 " & 2 "

(iii) 2 " & 2 others alike

(iv) all different.

(i) 3 E's & one of X, R, C, I, S can be done
in 5C_1 ways

(ii) 2 E's & two of X, R, C, I, S can be done
in 5C_2 ways

2 S's & two of E, X, R, C, I can be done
in 5C_2 ways

(iii) 2 E's & 2 S's can be done in only
1 way

(iv) 4 letters from E, X, R, C, I, S can be done in
 6C_4 ways

No. of different arrangements of each classification is

$$(i) {}^5C_1 \times \frac{4!}{3!} = 20$$

$$(ii) ({}^5C_2 + {}^5C_3) \times \frac{4!}{2!} = 240$$

(3)

$$(iii) 1 \times \frac{4!}{2! \cdot 2!} = 6$$

$$(iv) {}^6C_4 \times 4! = 360$$

$\therefore$ Total no. of different arrangements is 626.

Question 4.

a) If $P(x)$ is divided by $(x-3)(x-4)$ then

$$P(x) = (x-3)(x-4) \cdot Q(x) + R(x)$$

where $R(x) = ax+b$ (deg of $R(x) <$ deg of divisor)

$$\text{Now } P(3) = 10$$

$$\therefore 10 = 3a+b \quad \text{--- (1)}$$

$$\text{if } P(4) = 13$$

$$\therefore 13 = 4a+b \quad \text{--- (2)}$$

$$\textcircled{2} - \textcircled{1}$$

$$3 = a$$

$$b = 1$$

$\therefore$ the remainder is $3x+1$

$$\text{b) (i) Let } y = \frac{1}{x}$$

since $x = \alpha, \beta, \gamma$

$$y = \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$$

$$\therefore x = \frac{1}{y}$$

$$\therefore \left(\frac{1}{y}\right)^3 - 7\left(\frac{1}{y}\right)^2 - 7 = 0$$

$$\frac{1}{y^3} - \frac{7}{y^2} - 7 = 0$$

$$1 - 7y - 7y^3 = 0$$

i.e

$$0 = 7y^3 + 7y - 1$$

$\therefore$ the required eqⁿ is $7x^3 + 7x - 1 = 0$

$$\text{(ii) Let } y = x^2$$

since $x = \alpha, \beta, \gamma$

$$y = \alpha^2, \beta^2, \gamma^2$$

$$x = \sqrt{y}$$

$$\therefore (\sqrt{y})^3 - 7(\sqrt{y})^2 - 7 = 0$$

$$y^{\frac{3}{2}} - 7y - 7 = 0$$

$$y^{\frac{3}{2}} = 7y + 7$$

Square both sides

$$y^3 = (7y + 7)^2$$

$$y^3 = 49y^2 + 98y + 49$$

(2)

$$y^3 - 49y^2 - 98y + 49 = 0$$

∴ The required eqⁿ is $x^3 - 49x^2 - 98x - 49 = 0$.

c) (1) $\frac{2}{x^3 + 2x} = \frac{2}{2(x^2 + 2)}$

Let $\frac{2}{2(x^2 + 2)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 2}$

∴ $2 = A(x^2 + 2) + x(Bx + C)$

When $x = 0$, $2 = 2A$
 $A = 1$

When $x = 1$, $2 = 3A + B + C$
 $2 = 3 + B + C$

$B + C = -1$ — (1)

When $x = -1$, $2 = 3A + B - C$
 $2 = 3 + B - C$

$B - C = -1$ — (2)

(1) + (2)

$2B = -2$

$B = -1$

∴ $C = 0$

∴ $\frac{2}{x^3 + 2x} = \frac{1}{x} - \frac{x}{x^2 + 2}$

(2)

$$\begin{aligned}
 (11) \int_1^2 \frac{2}{x^3+2x} dx &= \int_1^2 \frac{1}{x} - \frac{x}{x^3+2} dx \\
 &= \left[\ln x - \frac{1}{2} \ln(x^3+2) \right]_1^2 \\
 &= \left(\ln 2 - \frac{1}{2} \ln 6 \right) - \left(\ln 1 - \frac{1}{2} \ln 3 \right) \\
 &= \ln 2 - \ln \sqrt{6} + \ln \sqrt{3} \\
 &= \ln \left(\frac{2\sqrt{3}}{\sqrt{6}} \right) \\
 &= \ln \left(\frac{2}{\sqrt{2}} \right) \\
 &= \ln(\sqrt{2}) \\
 &= \frac{1}{2} \ln 2
 \end{aligned}$$

(2)

d) (i) If $1+i$ is a root of the eqⁿ we may sub $1+i$ into the eqⁿ.

$$\begin{aligned}
 \text{Now } (1+i)^2 &= 1 + 2i + i^2 \\
 &= 2i
 \end{aligned}$$

$$\begin{aligned}
 (1+i)^3 &= 2i(1+i) \\
 &= -2 + 2i
 \end{aligned}$$

$$\begin{aligned}
 (1+i)^4 &= (2i)^2 \\
 &= -4
 \end{aligned}$$

$$\therefore \text{If } (1+i)^4 + p(1+i)^3 + q(1+i) + r = 0$$

$$-4 + p(-2+2i) + q(1+i) + r = 0$$

$$\text{But } \sum \alpha = -p \quad \text{and } \sum \alpha\beta\gamma\delta = r$$

$$\text{Hence } -p = b + r$$

$$r = -p - b$$

$$\therefore -4 + p(-2+2i) + q(1+i) - p - q = 0$$

$$-4 - 2p + 2pi + q + qi - p - q = 0$$

$$-10 - 3p + q + i(2p+q) = 0 + 0i$$

$$\therefore 2p+q = 0 \quad \text{--- (1)}$$

$$-3p+q = 10 \quad \text{--- (2)}$$

$$(1) - (2)$$

$$5p = -10$$

$$p = -2$$

$$\therefore q = 4$$

$$\therefore r = -4.$$

(3)

(iii) Since $(1+i)$ is a root $\therefore p, q, r$ are real
 $(1-i)$ is also a root

Let the roots be $\alpha, \beta, 1+i, 1-i$

$$\text{Now } \sum \alpha = \alpha + \beta + 2$$

$$\therefore \alpha + \beta + 2 = -p$$

$$\alpha + \beta + 2 = 2$$

$$\alpha = -\beta \quad \text{--- (1)}$$

$$\text{Also } \sum \alpha\beta\gamma\delta = \alpha\beta(1-i)(1+i)$$

$$= \alpha\beta \times 2$$

$$= 2\alpha\beta$$

(2)

$$\therefore 2\alpha\beta = r$$

$$2\alpha\beta = -4$$

$$\alpha\beta = -2 \quad \text{--- (2)}$$

Sub (1) into (2)

$$-\beta^2 = -2$$

$$\beta^2 = 2$$

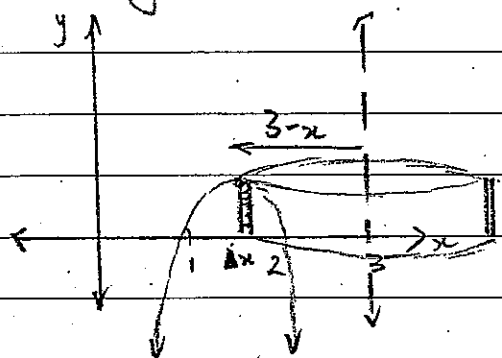
$$\beta = \pm\sqrt{2}$$

$$\alpha = \pm\sqrt{2}$$

Roots are $\sqrt{2}, \sqrt{2}, (1+i), (1-i)$

QUESTION 5.

a) If $y = -2 + 3x - x^2$
 $y = (3-x)(x-1)$



$$\Delta V = 2\pi xy \cdot \Delta x$$

$$= 2\pi (3-x)(-2+3x-x^2) \cdot \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n 2\pi (3-x)(-2+3x-x^2) \Delta x$$

$$= 2\pi \int_1^3 (3-x)(-2+3x-x^2) dx$$

$$= 2\pi \int_1^3 (-6 + 9x - 3x^2 + 2x - 3x^2 + x^3) dx$$

$$= 2\pi \int_1^3 (x^3 - 6x^2 + 11x - 6) dx$$

$$= 2\pi \left[\frac{x^4}{4} - 2x^3 + \frac{11}{2}x^2 - 6x \right]_1^3$$

$$= 2\pi \left[\left(\frac{81}{4} - 2(27) + \frac{11}{2}(9) - 6(3) \right) - \left(\frac{1}{4} - 2 + \frac{11}{2} - 6 \right) \right]$$

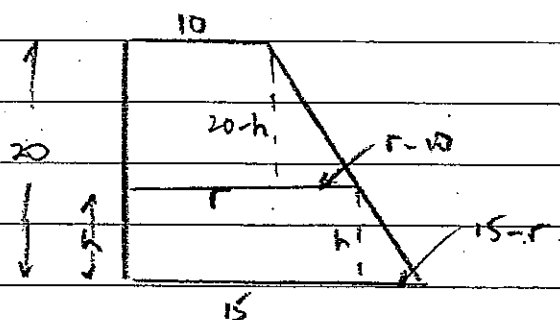
$$= 2\pi \left[(4 - 16 + 22 - 12) - \left(\frac{1}{4} - 2 + \frac{11}{2} - 6 \right) \right]$$

$$= 2\pi \times \frac{1}{4}$$

$$= \frac{\pi}{2} \text{ units}^3$$

5

b) (i)



Since the Δ s are similar

$$\frac{r-10}{15-r} = \frac{20-h}{h}$$

$$hr - 10h = (15-r)(20-h)$$

$$hr - 10h = 300 - 15h - 20r + rh$$

$$20r = 300 - 5h$$

$$r = \frac{300 - 5h}{20}$$

(2)

$$r = 15 - \frac{h}{4}$$

(ii) Area of cross-section is πr^2

$$\therefore A = \pi \left(15 - \frac{h}{4} \right)^2$$

$$\Delta V = A \cdot \Delta h$$

$$= \pi \left(15 - \frac{h}{4} \right)^2 \cdot \Delta h$$

$$V = \lim_{\Delta h \rightarrow 0} \sum_0^{20} \pi \left(15 - \frac{h}{4} \right)^2 \cdot \Delta h$$

$$= \int_0^{20} \pi \left(15 - \frac{h}{4} \right)^2 dh$$

$$= \pi \left[\frac{\left(15 - \frac{h}{4} \right)^3}{3 \times -\frac{1}{4}} \right]_0^{20}$$

$$= \pi \left[\frac{(15-5)^3}{-\frac{3}{4}} - \frac{(15-0)^3}{-\frac{3}{4}} \right]$$

$$= \pi \times 3166 \frac{2}{3}$$

(3)

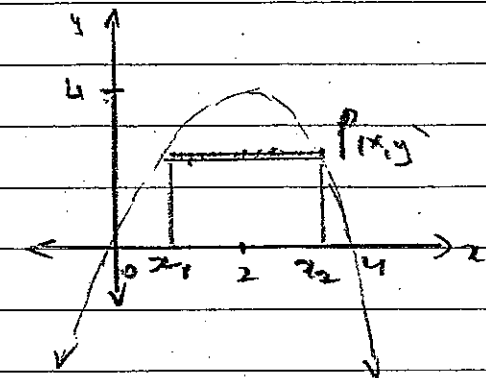
$$= \frac{9500\pi}{3} \text{ units}^3$$

$$c) (i) \text{ If } 4x - x^2 = 0$$

$$x(4-x) = 0$$

$$x = 0 \text{ or } 4$$

$$\text{When } x = 2, y = 4 - 4 = 0$$



The annulus has inner radius x_1 , outer radius x_2

$$\text{Area of annulus} = \pi (x_2^2 - x_1^2)$$

$$A = \pi (x_2 + x_1)(x_2 - x_1)$$

where x_1 and x_2 are the roots of $y = 4x - x^2$ for various values of y

$$\text{Since } y = 4x - x^2$$

$$x^2 - 4x + y = 0$$

$\therefore x_1, x_2$ are the roots

$$x_1 + x_2 = \frac{-b}{a} = 4, \quad x_1 x_2 = \frac{c}{a} = y$$

$$\begin{aligned} \text{Now } (x_2 - x_1)^2 &= (x_2 + x_1)^2 - 4x_1 x_2 \\ &= 4^2 - 4y \\ &= 16 - 4y \end{aligned}$$

$$x_2 - x_1 = \sqrt{16 - 4y}$$

$$\begin{aligned} \therefore A &= \pi (4) (\sqrt{16 - 4y}) \\ &= \pi [4\sqrt{16 - 4y}] \end{aligned}$$

(3)

$$(ii) \Delta V = \pi [4\sqrt{16 - 4y}] \cdot \Delta y$$

$$V = \lim_{\Delta y \rightarrow 0} \sum_0^4 \pi [4\sqrt{16 - 4y}] \Delta y$$

$$= 4\pi \int_0^4 \sqrt{16 - 4y} \, dy$$

$$= 4\pi \int_0^4 2\sqrt{4 - y} \, dy$$

$$= 8\pi \int_0^4 (4 - y)^{\frac{1}{2}} \, dy$$

$$= 8\pi \left[\frac{(4-y)^{3/2}}{\frac{3}{2}x-1} \right]_0^4$$

$$= 8\pi \left[-\frac{2}{3}(4-y)^{3/2} \right]_0^4$$

$$= 8\pi \left[-\frac{2}{3} \times 0 + \frac{2}{3} \times 4^{3/2} \right]$$

(2)

$$= 8\pi \times \frac{2}{3} \times 8$$

$$= \frac{128\pi}{3} \text{ units}^3$$

QUESTION 6.

$$a) (i) \text{ If } \frac{x^2}{100} + \frac{y^2}{64} = 1$$

$$a = 10, \quad b = 8.$$

$$b^2 = a^2(1 - e^2)$$

$$64 = 100(1 - e^2)$$

$$\frac{64}{100} = 1 - e^2$$

$$e^2 = 1 - \frac{64}{100}$$

(1)

$$e^2 = \frac{36}{100}$$

$$e = \frac{3}{5}, \quad e > 0.$$

(ii) Foci are $(\pm ae, 0)$

$$ae = 10 \times \frac{3}{5}$$

$$= 6$$

Foci are $(\pm 6, 0)$

Directrices are $x = \pm \frac{a}{e}$

(2)

$$\frac{a}{e} = \frac{10}{\frac{3}{5}}$$

$$= \frac{50}{3}$$

Directrices are $x = \pm \frac{50}{3}$

$$(iii) \text{ If } \frac{x^2}{100} + \frac{y^2}{64} = 0$$

$$\frac{2x}{100} + \frac{2y}{64} \cdot y' = 0$$

$$\frac{2y}{64} \cdot y' = \frac{-2x}{100}$$

$$y' = \frac{-x}{\frac{100}{64}}$$

$$= -\frac{64x}{100y}$$

At $P(x_0, y_0)$, $y' = -\frac{64}{100} \frac{x_0}{y_0}$

∴ The eqn of the tangent at P is

$$y - y_0 = \frac{-64}{100} \frac{x_0}{y_0} (x - x_0)$$

$$\frac{y_0(y - y_0)}{-64} = \frac{x_0(x - x_0)}{100}$$

(3)

$$-\frac{y_0 y}{64} + \frac{y_0^2}{64} = \frac{x_0 x}{100} - \frac{x_0^2}{100}$$

$$\frac{x_0^2}{100} + \frac{y_0^2}{64} = \frac{x_0 x}{100} + \frac{y_0 y}{64}$$

∴ $1 = \frac{x_0 x}{100} + \frac{y_0 y}{64}$ since (x_0, y_0) lies on E

i.e. $\frac{x_0 x}{100} + \frac{y_0 y}{64} = 1$

b) i) Since the focus is $(ae, 0)$

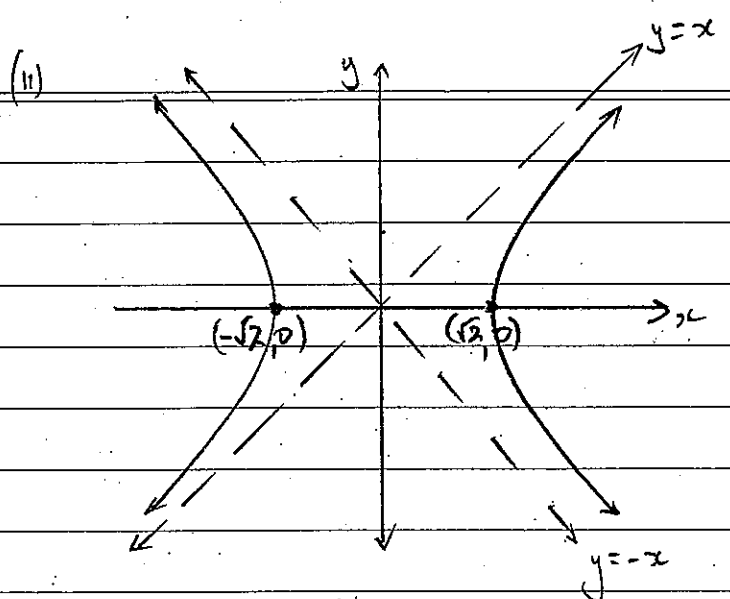
$$ae = 2$$

$$a \cdot \sqrt{2} = 2$$

$$a = \sqrt{2}$$

(1)

Hence the eqn is $\frac{x^2}{2} - \frac{y^2}{2} = 1$



(iii) $\frac{x^2}{2} - \frac{y^2}{2} = 1$

$$x^2 - y^2 = 2$$

$$\therefore 2x - 2y \cdot y' = 0$$

$$y' = \frac{2x}{2y}$$

$$y' = \frac{x}{y}$$

at $P(\sqrt{2} \sec \theta, \sqrt{2} \tan \theta)$ $y' = \frac{\sqrt{2} \sec \theta}{\sqrt{2} \tan \theta}$

$$= \frac{\frac{1}{\cos \theta}}{\frac{\sin \theta}{\cos \theta}}$$

$$= \frac{1}{\sin \theta}$$

$\therefore$ the gradient of the normal is $-\sin \theta$

Hence the eqⁿ is

$$y - \sqrt{2} \tan \theta = -\sin \theta (x - \sqrt{2} \sec \theta)$$

$$y - \sqrt{2} \tan \theta = -\sin \theta x + \sqrt{2} \sin \theta \sec \theta$$

$$y - \sqrt{2} \tan \theta = -\sin \theta x + \sqrt{2} \tan \theta$$

$$\sin \theta x + y = 2\sqrt{2} \tan \theta$$

$\div$ throughout by $\cos \theta$

$$\frac{\sin \theta}{\cos \theta} x + \frac{y}{\cos \theta} = 2\sqrt{2} \frac{\tan \theta}{\cos \theta}$$

$$\tan \theta \cdot x + \sec \theta \cdot y = 2\sqrt{2} \tan \theta \sec \theta$$

$$\text{i.e. } x \tan \theta + y \sec \theta = 2\sqrt{2} \sec \theta \tan \theta \quad (3)$$

$$(ix) \text{ When } y=0, \quad x \tan \theta = 2\sqrt{2} \sec \theta \tan \theta$$

$$\therefore x = 2\sqrt{2} \sec \theta$$

$$Q \text{ is } (2\sqrt{2} \sec \theta, 0)$$

$$\text{When } x=0, \quad y \sec \theta = 2\sqrt{2} \sec \theta \tan \theta$$

$$\therefore y = 2\sqrt{2} \tan \theta$$

$$P \text{ is } (0, 2\sqrt{2} \tan \theta)$$

$$\therefore T \text{ is } (2\sqrt{2} \sec \theta, 2\sqrt{2} \tan \theta)$$

$$\text{If } x = 2\sqrt{2} \sec \theta, \quad \sec \theta = \frac{x}{2\sqrt{2}} \quad (4)$$

$$\text{If } y = 2\sqrt{2} \tan \theta, \quad \tan \theta = \frac{y}{2\sqrt{2}}$$

$$\text{Now } 1 + \tan^2 \theta = \sec^2 \theta$$

$$\therefore 1 + \frac{y^2}{8} = \frac{x^2}{8}$$

$$\text{i.e. } \frac{x^2}{8} - \frac{y^2}{8} = 1$$

$$a = 2\sqrt{2}, \quad e = \sqrt{2}$$

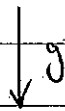
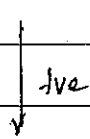
$$\therefore ae = 4$$

$$\frac{a}{e} = \frac{2\sqrt{2}}{\sqrt{2}} = 2$$

Thus the locus of T is a rectangular hyperbola with vertices $(\pm 2\sqrt{2}, 0)$, foci $(\pm 4, 0)$ and directrices $x = \pm 2$.

QUESTION 7

a) (i)



$$F = ma, \quad m = 1$$

$$\therefore \ddot{y} = g - kv \quad (1)$$

$$\text{or } \frac{dv}{dt} = g - kv$$

When $t=0$, $v=0$ since it is instantaneously at rest at max. height

$$(ii) \quad \frac{dv}{dt} = g - kv$$

$$\frac{dt}{dv} = \frac{1}{g - kv}$$

$$t = \int -\frac{1}{k} \cdot \frac{-k}{g - kv} dv$$

$$\therefore t = -\frac{1}{k} \ln(g - kv) + c$$

$$\text{If } t=0, v=0$$

$$0 = -\frac{1}{k} \ln(g) + c$$

$$c = \frac{1}{k} \ln g$$

$$\therefore t = -\frac{1}{k} \ln(g - kv) + \frac{1}{k} \ln g$$

$$t = \frac{1}{k} \ln \left(\frac{g}{g - kv} \right)$$

$$kt = \ln \left(\frac{g}{g - kv} \right)$$

$$e^{kt} = \frac{g}{g - kv}$$

$$e^{kt}(g - kv) = g$$

$$g - kv = \frac{g}{e^{kt}}$$

$$kv = g - ge^{-kt}$$

$$v = \frac{g}{k} (1 - e^{-kt}), \quad t \geq 0$$

(3)

$$(iii) \quad \frac{dr}{dt} = r \frac{dr}{dy}$$

$$\therefore r \frac{dr}{dy} = g - kv$$

$$\frac{dr}{dy} = \frac{g - kv}{r}$$

$$\frac{dy}{dr} = \frac{r}{g - kv}$$

$$= -\frac{1}{k} \left(\frac{-kv}{g - kv} \right)$$

$$= -\frac{1}{k} \left(\frac{g - kv - g}{g - kv} \right)$$

$$= -\frac{1}{k} \left(1 - \frac{g}{g - kv} \right)$$

$$y = \int -\frac{1}{k} \left(1 - \frac{g}{g - kv} \right) dr$$

$$y = -\frac{1}{k} \int 1 + \frac{g}{k} \cdot \frac{-k}{g - kv} dr$$

$$y = -\frac{1}{k} \left(r - \frac{g}{k} \ln(g - kv) \right) + C_1$$

$$-ky = r + \frac{g}{k} \ln(g - kv) + C_2$$

$$\text{If } v=0, y=0$$

$$0 = 0 + \frac{g}{k} \ln(g - 0) + C_2$$

$$C_2 = -\frac{g}{k} \ln g$$

$$\therefore -ky = r + \frac{g}{k} \ln(g - kv) - \frac{g}{k} \ln g$$

$$-ky = r + \frac{g}{k} \ln \left(\frac{g - kv}{g} \right)$$

$$\text{i.e. } \frac{g}{k} \ln \left(\frac{g - kv}{g} \right) + r = -kv$$

(iv) From (iii)

$$v + ky = -\frac{g}{k} \ln\left(\frac{g - kv}{g}\right)$$

$$\text{From (ii)} \quad v = \frac{g}{k} (1 - e^{-kt})$$

$$\therefore v + ky = -\frac{g}{k} \ln \left[\frac{g - k \cdot \frac{g}{k} (1 - e^{-kt})}{g} \right]$$

$$\frac{v + ky}{g} = -\frac{1}{k} \ln \left[\frac{g - g(1 - e^{-kt})}{g} \right]$$

$$\frac{v + ky}{g} = -\frac{1}{k} \ln (1 - 1 + e^{-kt})$$

$$\frac{v + ky}{g} = -\frac{1}{k} \ln (e^{-kt})$$

$$\frac{v + ky}{g} = -\frac{1}{k} \times -kt \quad (2)$$

$$\frac{v + ky}{g} = t$$

$$\text{i.e. } t = \frac{v + ky}{g}$$

(v) When the particle returns to ground level
 $y = h$

$$\therefore t = \frac{v + kh}{g}$$

$$= \frac{v}{g} + \frac{k}{g} \cdot \frac{1}{k} \left[u - \frac{g}{k} \ln \left(\frac{g + ku}{g} \right) \right]$$

$$= \frac{v}{g} + \frac{u}{g} - \frac{1}{k} \ln \left(\frac{g + ku}{g} \right)$$

∴ the total time T is given by

$$T = \frac{1}{k} \log_e \left(\frac{g+kU}{g} \right) + \frac{V}{g} + \frac{U}{g} - \frac{1}{k} \log_e \left(\frac{g+kU}{g} \right)$$

$$T = V + \frac{U}{g} \quad (1)$$

b) $I_n = \int_0^1 (x^2-1)^n dx$

(i) $I_0 = \int_0^1 1 dx$ (1)

$$= [x]_0^1$$

$$= 1$$

(ii) $I_n = \int_0^1 (x^2-1)^n dx$ $u = (x^2-1)^n \quad u' = 1$

$\therefore I_n = \left[x(x^2-1)^n \right]_0^1 - 2n \int_0^1 x^2 (x^2-1)^{n-1} dx$ $u' = n(x^2-1)^{n-1} \cdot 2x \quad v = x$

$$= 0 - 2n \int_0^1 [(x^2-1)+1] (x^2-1)^{n-1} dx$$

$$= 0 - 2n \int_0^1 (x^2-1)^n + (x^2-1)^{n-1} dx$$

$$= 0 - 2n (I_n + I_{n-1})$$

$$I_n + 2n I_n = -2n I_{n-1}$$

$$I_n (2n+1) = -2n I_{n-1}$$

$$I_n = \frac{-2n}{2n+1} I_{n-1}$$

(iii) $I_4 = \frac{-8}{10} I_3$

$$= \frac{-8}{10} \times \frac{-6}{8} I_2$$

$$= \frac{-8}{10} \times \frac{-6}{8} \times \frac{-4}{6} I_1$$

$$= \frac{-8}{10} \times \frac{-6}{8} \times \frac{-4}{6} \times \frac{-2}{4} I_0$$

$$= \frac{128}{315} I_0$$

QUESTION 8.

a) (i) $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$

$$\text{LHS } (\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4\cos^3 \theta i \sin \theta + 6\cos^2 \theta i^2 \sin^2 \theta + 4\cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta$$

$$\text{RHS} = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta + i(4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta)$$

Equate real & imaginary parts

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$\sin 4\theta = 4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta$$

$$\tan 4\theta = \frac{4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta}{\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

÷ throughout by $\cos^4 \theta$

$$\tan 4\theta = \frac{4\cancel{\cos^3 \theta} \sin \theta}{\cancel{\cos^4 \theta}} - \frac{4\cancel{\cos} \sin^3 \theta}{\cancel{\cos^3 \theta}}$$

$$= \frac{\cancel{\cos^4 \theta}}{\cancel{\cos^4 \theta}} - 6\frac{\cancel{\cos^2 \theta} \sin^2 \theta}{\cancel{\cos^2 \theta}} + \frac{\sin^4 \theta}{\cancel{\cos^4 \theta}}$$

$$\tan 4\theta = \frac{4\tan \theta - 4\tan^3 \theta}{1 - 6\tan^2 \theta + \tan^4 \theta}$$

If $t = \tan \theta$

(4)

$$\tan 4\theta = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$$

(ii) If $\tan 4\theta = 1$

$$1 = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$$

$$1 - 6t^2 + t^4 = 4t - 4t^3$$

$$\text{i.e. } t^4 + 4t^3 - 6t^2 - 4t + 1 = 0$$

Now when $\tan 4\theta = 1$

$$4\theta = n\pi + \frac{\pi}{4}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$\begin{aligned} 4\theta &= \frac{4n\pi + \pi}{4} \\ &= \pi \left(\frac{4n+1}{4} \right) \end{aligned}$$

3

$\therefore \theta = \frac{\pi(4n+1)}{16}$, $n = 0, \pm 1, -2$ give 4 distinct values

$$\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{-3\pi}{16}, \frac{-7\pi}{16}$$

$$\therefore t = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{-3\pi}{16}, \tan \frac{-7\pi}{16}$$

⇒ (i) Horizontally

$$\ddot{x} = 0$$

$$\dot{x} = V \cos \alpha$$

$$x = V \cos \alpha \cdot t$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + V \sin \alpha$$

$$y = -\frac{1}{2}gt^2 + V \sin \alpha \cdot t$$

$$\text{If } y = 0,$$

$$-\frac{1}{2}gt^2 + V \sin \alpha \cdot t = 0$$

$$t(-\frac{1}{2}gt + V \sin \alpha) = 0$$

$$t = 0 \quad \text{or} \quad \frac{2V \sin \alpha}{g}$$

$$\text{When } t = \frac{2V \sin \alpha}{g}$$

$$x = V \cos \alpha \cdot \frac{2V \sin \alpha}{g}$$

$$\therefore x = \frac{v^2 \sin 2\alpha}{g}$$

Max range occurs when $\sin 2\alpha = 1$

$$\text{i.e. } 2\alpha = \frac{\pi}{2}$$

$$\alpha = \frac{\pi}{4}$$

(4)

2. Max range is $\frac{v^2}{g}$

(ii) Let the new velocity be W

$$x = W \cos \alpha \cdot t, \quad y = -\frac{1}{2}gt^2 + W \sin \alpha \cdot t$$

$$\text{When } \alpha = \frac{\pi}{4}, \quad x = \frac{v^2}{g}$$

$$\therefore x = W \cos \frac{\pi}{4} \cdot t$$

$$\text{So } \frac{v^2}{g} = W \cdot \frac{1}{\sqrt{2}} \cdot t$$

$$\therefore t = \frac{v^2 \sqrt{2}}{gW}$$

$$\text{When } t = \frac{v^2 \sqrt{2}}{gW}$$

$$y = -\frac{1}{2}gt^2 + W \sin \alpha \cdot t$$

$$= -\frac{g}{2} \cdot \frac{v^4 \cdot 2}{g^2 W^2} + \frac{W}{\sqrt{2}} \cdot \frac{v^2 \sqrt{2}}{gW}$$

$$\therefore h = -\frac{v^4}{gW^2} + \frac{v^2}{g}$$